Functional data analysis

4. Exploratory Data Analysis

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Lecture notes, 2026

Objectives:

- *to discuss sample mean and sample covariance and other descriptive parameters of functional data;*
- *to introduce notion of depth for functional data;*
- *to discuss outliers;*
- *to introduce the functional box-plot and other visualization tools for functional data;*
- *registration.*

Outline of the lecture

- Point-wise exploratory data analysis
- Sample mean and sample covariance functions
- Principal component functions
- Functional Depth
- Functional Outlier Detection
- Functional boxplots
- Visualization tools
- Curves alignment

Point-wise exploratory data analysis

- Consider functional data sample $x_1, \ldots, x_n$, where each function is defined on the interval $\mathcal{T} = [0, T]$,

$$x_i = (x_i(t), t \in \mathcal{T}), \quad i = 1, \ldots, n.$$

- For each $t \in \mathcal{T}$ we have univariate data set

$$x_1(t), \cdots, x_n(t)$$

for which we can do both quantitative (numerical) and qualitative (graphical) summaries. This constitute point-wise exploratory data analysis.

- Point-wise quantitative summaries include:
  - sample mean,
  - sample median,
  - sample variance,
  - mean absolute deviation,
  - quartiles, IQR, outliers, skewness.
- Point-wise graphical summaries include:
  - histogram,
  - bin width,
  - bin centres,
  - boxplot.
- In univariate exploratory data analysis usually one assumes that the sample corresponds to random sample of independent random variables. For functional data however the main interest is to determine intra-time ties of the functions.

Sample mean and sample covariance functions

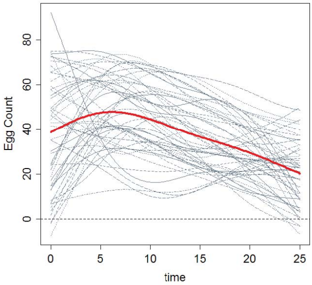
- We now assume that the raw data have been converted to functional objects by a suitable basis expansion or some other smoothing method.
- So we have a sample consisting of n curves

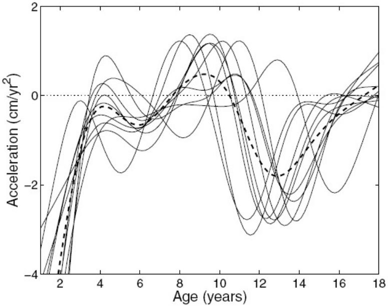
$$x_1, x_2, \ldots, x_n,$$

where for each $i = 1, \ldots, n$, $x_i = (x_i(t),\ t \in \mathcal{T})$.
- We view each curve x_i as a realization of a random function $X_i = \{X_i(t), t \in \mathcal{T})$.

- *Sample mean function* is defined as function $\overline{x}_n = (\overline{x}_n(t), t \in \mathcal{T})$, where

$$\overline{x}_n(t) = \frac{1}{n}\sum_{i=1}^{n} x_i(t).$$



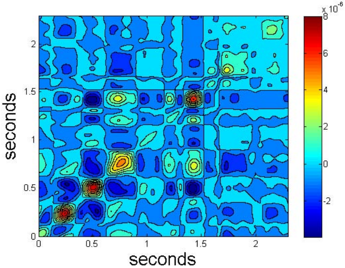


- *Sample variance function* is defined as function $\widehat{\sigma}_x^2 = (\widehat{\sigma}_x^2(t), t \in \mathcal{T})$, where

$$\widehat{\sigma}_x^2(t) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i(t) - \overline{x}_n(t))^2.$$

- *Sample standard deviation function* is defined by $\mathrm{sd}_x = \sqrt{\widehat{\sigma}_x^2}$.
- The sample standard deviation gives no information on how the values of the curves at point t relate to those at point s.
- Such information one obtains from the *sample covariance function* defined as function $\widehat{c}_x = (\widehat{c}_x(t,s), t, s \in \mathcal{T})$, where

$$\widehat{c}(t,s) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i(t) - \overline{x}_n(t))(x_i(s) - \overline{x}_n(s)).$$


x 10^-6
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4
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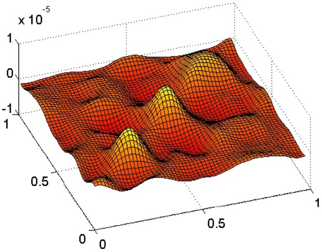
seconds


x 10^-5
1
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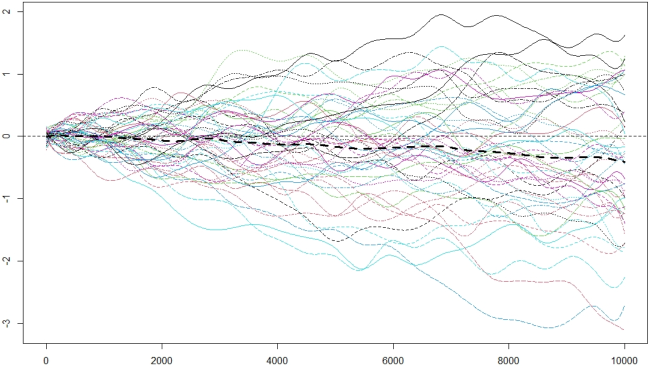
-1
1
0.5
0
0
0.5
1

Example: Gaussian random walk

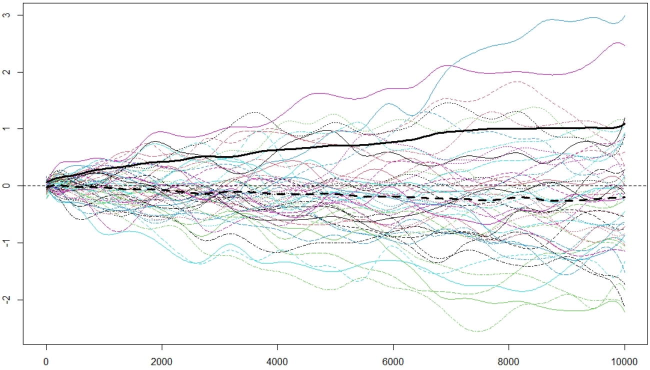
```
N=50
W.mat=matrix(0, ncol=N, nrow=10000)
for(n in 1:N)W.mat[, n]=cumsum(rnorm(10000))/100
B25.basis=create.bspline.basis(rangeval=c(0,10000),
nbasis=25)
W.fd=smooth.basis(y=W.mat, fdParobj=B25.basis)
plot(W.fd, ylab="", xlab="",col="gray",lty=1)
W.mean=mean.fd(W.fd$fd)
lines(W.mean, lty=2, lwd=3)
```



2
1
0
-1
-2
-3
0
2000
4000
6000
8000
10000

Example: Gaussian random walk

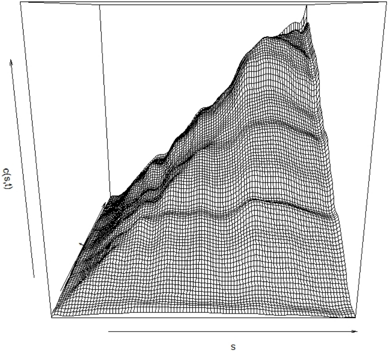
```
N=50
W.mat=matrix(0, ncol=N, nrow=10000)
for(n in 1:N)W.mat[,n]=cumsum(rnorm(10000))/100
B25.basis=create.bspline.basis(rangeval=c(0,10000),
nbasis=25)
W.fd=smooth.basis(y=W.mat, fdParobj=B25.basis)
plot(W.fd, ylab="", xlab="",col="gray",lty=1)
W.mean=mean(W.fd$fd)
W.sd=std.fd(W.fd$fd)
lines(W.sd, lwd=3); lines(W.mean, lty=2, lwd=3)
```

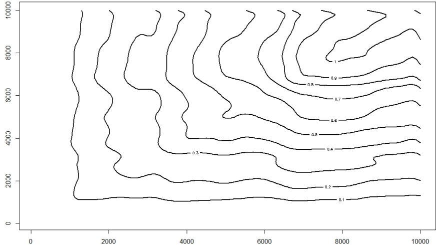


3
2
1
0
-1
-2
0
2000
4000
6000
8000
10000

Example: Gaussian random walk

```
W.cov=var.fd(W.fd$fd)   {$fd extracts function values}
grid=(1:100)*100
W.cov.mat=eval.bifd(grid, grid, W.cov)
persp(grid, grid, W.cov.mat, xlab="s", ylab="t", zlab="c(s,t)")
contour(grid, grid, W.cov.mat, lwd=2)
```


c(s,t)
s


10000
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6000
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0.5
0.6
0.7
0.8
0.9
1

Principal component functions

- One of the most useful and often used tools in functional data analysis is the *principal component analysis*.
- *Estimated functional principal components*, EFPC's, are related to the sample covariance function $\widehat{c}(t, s)$, as will be explained later. Here we only explain how to compute and interpret them.
- The idea of the estimated functional principal component expansion is to find functions $\widehat{v}_j(t)$ such that the centred functions $x_i(t) - \overline{x}_n(t)$ are represented as

$$x_i(t) - \overline{x}_n(t) \approx \sum_{j=1}^{p} \widehat{z}_{ij} \widehat{v}_j(t), \quad t \in \mathcal{T},$$

  with p much smaller than the number of basis functions used to in reconstruction of $x_i(t)$.

- The estimated functional principal components (EFPC) $\widehat{v}_j$ are computed from the observed functions $x_1, x_2, \ldots, x_n$ after converting them to functional objects.
- `pca.fd(fdobj, nharm = 2, harmfdPar=fdPar(fdobj),`
  `        centerfns = TRUE)`
- Arguments:

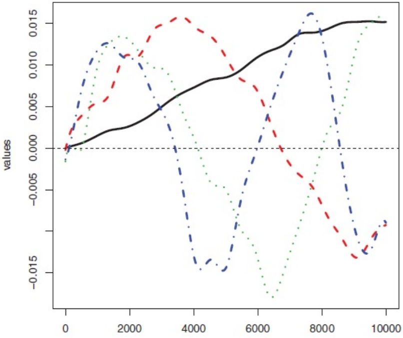
  `fdobj` - a functional data object.

  `nharm` - the number of harmonics or principal components to compute.

  `harmfdPar` - a functional parameter object that defines the harmonic or principal component functions to be estimated.

  `centerfns` - a logical value: if TRUE, subtract the mean function from each function before computing principal components.

```
W.pca = pca.fd(W.fd$fd, nharm=4)
plot(W.pca$harmonics, lwd=3)
```



The first four ESPC
of the 50 random walks

- W.pca$varprop

  0.80513155 0.09509154 0.04099496

- The first EFPC, $\widehat{v}_1(t)$, shows the most pronounced pattern of the deviation from the mean function of a randomly selected trajectory.
- For each curve $x_i(t)$, the coefficient $\widehat{z}_{i1}$ quantifies the contribution of $\widehat{v}_1(t)$ to its shape.
- The coefficient $\widehat{z}_{ij}$ is called the score of $x_i(t)$ with respect to $\widehat{v}_j(t)$.
- The second EFPC, shows the second most important mode of departure from the mean functions of the curves under investigation.
- The EFPC's $\widehat{v}_j$ are orthonormal, in the sense that

$$\int_{\mathcal{T}} \widehat{v}_j(t)\widehat{v}_k(t)dt = \begin{cases} 0 & \text{if } \; j \neq k; \\ 1 & \text{if } \; j = k. \end{cases}$$

- We will see that the total variability of a sample of curves about the sample mean function, can be decomposed into the sum of variabilities explained by each EFPC.

Canadian weather PCA I

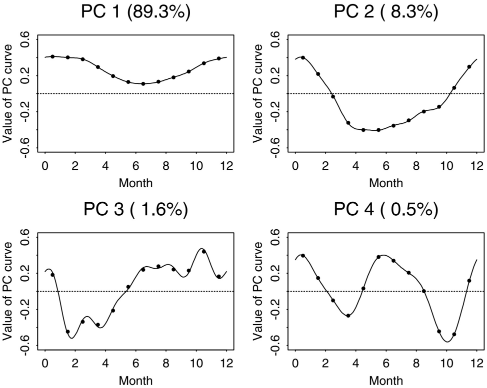


Figure 8.1. The first four principal component curves of the Canadian temperature data estimated by two techniques. The points are the estimates from the discretization approach, and the curves are the estimates from the expansion of the data in terms of a 12-term Fourier series. The percentages indicate the amount of total variation accounted for by each principal component.

Canadian weather PCA II

First principal component

1. Consider as an illustration in the upper left panel in Figure 8.1. This displays the weight function ξ_1 for the Canadian temperature data after the mean across all 35 weather stations has been removed from each station's monthly temperature record.
2. Although ξ_1 is positive throughout the year, the weight placed on the winter temperatures is about four times that placed on summer temperatures. This means that the greatest variability between weather stations will be found by heavily weighting winter temperatures, with only a light contribution from the summer months. **Canadian weather is most variable in the wintertime, in short.**

Canadian weather PCA III

3. Moreover, the percentage 89.3% at the top of the panel indicates that this type of variation strongly dominates all other types of variation. Weather stations for which the score f_{i1} is high will have much warmer than average winters combined with warm summers, and the two highest scores are in fact assigned to Vancouver and Victoria on the Pacific Coast. To no one's surprise, the largest negative score goes to Resolute in the High Arctic

Canadian weather PCA IV

Second principal component

- The weight function ξ_2 for the temperature data is displayed in the upper right panel of Figure 8.1. Because it must be orthogonal to ξ_1, we cannot expect that it will define a mode of variation in the temperature functions that will be as important as the first.
- In fact, this second mode accounts for only 8.3% of the total variation, and consists of a positive contribution for the winter months and a negative contribution for the summer months, therefore corresponding to a measure of uniformity of temperature through the year.

Canadian weather PCA V

- On this component, one of the highest scores f_{i2} goes to Prince Rupert, also on the Pacific coast, for which there is comparatively low discrepancy between winter and summer. Prairie stations such as Winnipeg, on the other hand, have hot summers and very cold winters, and receive large negative second component scores.

Canadian weather PCA VI

Third and fourth principal component

- The third and fourth components account for small proportions of the variation, since they are required to be orthogonal to the first two as well as to each other. At this point they are difficult to interpret

Plotting components as perturbation of means I

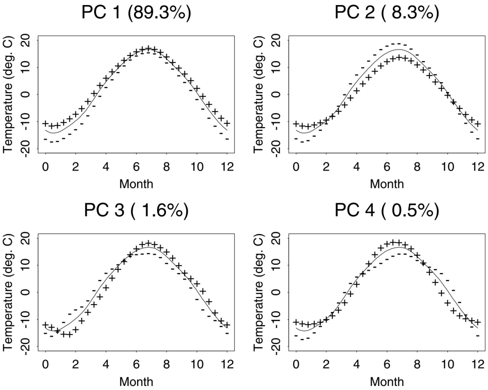


Figure 8.2. The mean temperature curves and the effects of adding (+) and subtracting (−) a suitable multiple of each PC curve.

Plotting components as perturbation of means II

1. We examine plots of the overall mean function and the functions obtained by adding and subtracting a suitable multiple of the principal component function in question.
2. Figure 8.2 shows such a plot for the temperature data. In each case, the solid curve is the overall mean temperature, and the dotted and dashed curves show the effects of adding and subtracting a multiple of each principal component curve.
3. This considerably clarifies the effects of the first two components.
4. We can now see that the third principal component corresponds to a time shift effect combined with an overall increase in temperature and in range between winter and summer.
5. The fourth corresponds to an effect whereby the onset of spring is later and autumn ends earlier.

VARIMAX rotation I

1. Varimax rotation is a statistical technique used at one level of factor analysis as an attempt to clarify the relationship among factors.
2. Generally, the process involves adjusting the coordinates of data that result from a principal components analysis. The adjustment, or rotation, is intended to maximize the variance shared among items. By maximizing the shared variance, results more discretely represent how data correlate with each principal component.
3. To maximize the variance generally means to increase the squared correlation of items related to one factor, while decreasing the correlation on any other factor. In other words, the varimax rotation simplifies the loadings of items by removing the middle ground and more specifically identifying the factor upon which data load.

VARIMAX rotation II

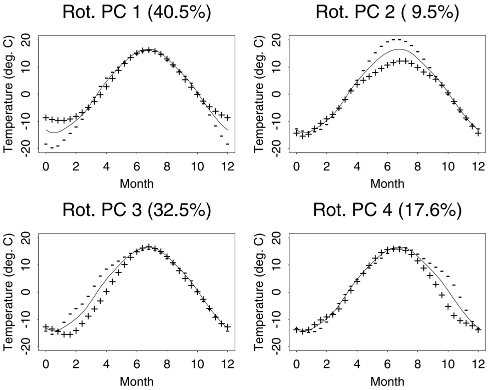


Figure 8.5. Weight functions rotated by applying the VARIMAX rotation criterion to weight function values, and plotted as positive and negative perturbations of the mean function.

VARIMAX rotation III

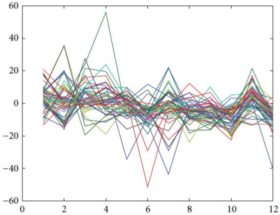
1. Figure 8.5 displays the VARIMAX rotation of the four principal components for the temperature data.
2. Collectively, the rotated functions still account for a total of 99.7% of the variation, but they divide this variation in different proportions.
3. The result is four functions that account for local variation in the winter, summer, spring and autumn, respectively. Not only are these functions much easier to interpret, but we see something new: although winter variation remains extremely important, now spring variation is clearly almost as important, about twice as important as autumn variation and over three times as important as summer variation.

Example: Shanghai Stock Exchange 50 Index

Following the research article:

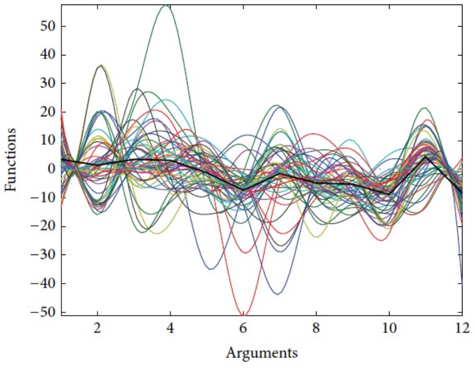
Zhiliang Wang, Yalin Sun, and Peng Li, Functional Principal Components Analysis of Shanghai Stock Exchange 50 Index, Hindawi Publishing Corporation Discrete Dynamics in Nature and Society Volume 2014, Article ID 365204, 7 pages.

- The main purpose of this paper is to explore the principle components of Shanghai stock exchange 50 index by means of functional principal component analysis (FPCA).



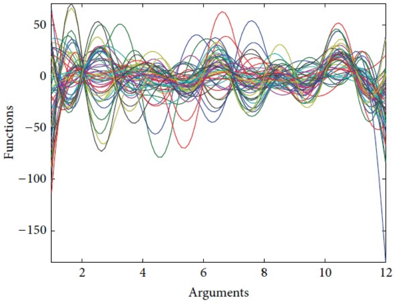
The monthly rate of return of 50 stocks.

Shanghai 50 datasets were converted to functional form:

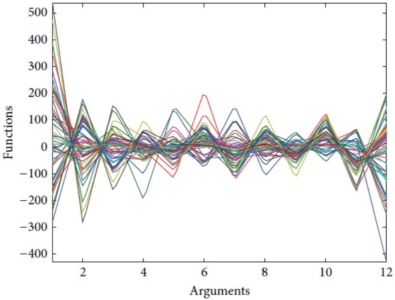


The functional form of 50 stocks. Note: the black bold line is the mean function.

Some curves are high (with good investment return) and some curves are low (with not so good investment return). It seems that that the variation from curve to curve can be explained at the level of certain derivatives.

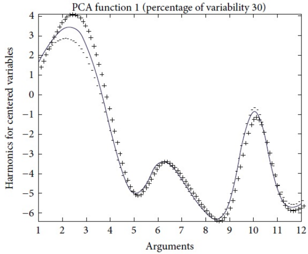


The first derivative curves.

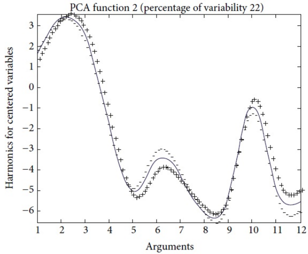


The second derivative curves.

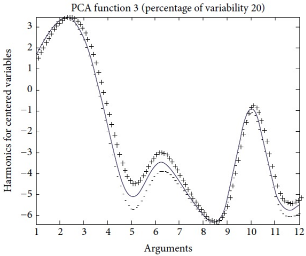
The fact that derivatives are of interest is further reason to think of the records as functions, rather than vectors of observations in discrete time.



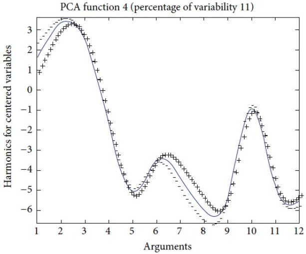
The first principal component function.



The second principal component function.



The third principal component function.



The fourth principal component function.

Shanghai 50 data: first four principal components

- "We can see that the first principal component function, which accounts for 30.22 percent of the variation, has always had an obvious positive effect on the mean function between February and April 2011. In fact, the concept of high-speed rail provoked the Chinese financial market vigorously during that period. Therefore, we can reasonably believe that the first principal component represents the speculation boom."
- "The second principal component function, which accounts for 22.07 percent of the variance, is seen to pick up the influence of tighten monetary policy to control the excessive price rises, especially the price of real estate. With the stock price getting lower and lower, more and more investors believe the current price is worth taking risk, which forms some power of buying driving price briefly rebounded."

- „The third principal component, which accounts for 19.91 percent of the variation, is believed to be the representative of the influence. With the excess drop in price, a growing number of blue chips are underestimated and the investment value will promote a price return."
- The effect is summarized to the fourth principal component, which accounts for 11.02 percent of the variation.

Conclusion:

FPCA attempts to find the dominant modes of variation around an overall trend function and is thus a key technique in functional data analysis. As we described before, modern data analysis has benefited and will continue to benefit greatly from the development of functional data analysis.

Life Course Data in Criminology I

James O. Ramsay, Bernard W. Silverman *Applied Functional Data Analysis: Methods and Case Studies*, Springer, 2002.

1. An important question in criminology is the study of the way that people's level of criminal activity varies through their lives.
2. Can it be said that there are "career criminals" of different kinds?
3. Are there particular patterns of persistence in the levels of crimes committed by individuals?
4. We concentrate on a single set of data giving the numbers of arrests of 413 men over a 25-year period in each of their lives, from age 11 to age 35.
5. The approach is to represent the criminal record of each subject by a single function of time, and then to use these functions for detailed analysis.

First steps in a functional approach

Life Course Data in Criminology II

1. We construct for each individual a function of time that represents his level of criminal activity. A simple approach would be to interpolate the raw numbers of arrests in each year, but because of the skewness of the annual counts this would give inordinate weight to high values in the original data.
2. In order to stabilize the variability somewhat, we start by taking the square root of the number of arrests each year.
3. If the numbers of arrests are Poisson counts, then the square root is the standard variance-stabilizing transformation.

Functional mean

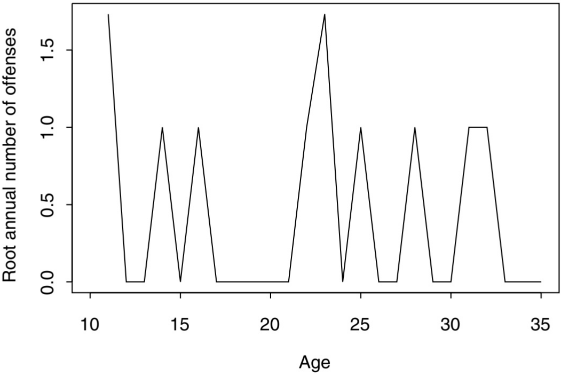


Figure 2.4. Linear interpolant of the square roots of the counts shown in Figure 2.2.

Life Course Data in Criminology IV

1. We now throw away the original points and regard this function as a whole as being the datum for this individual.
2. We denote by $Y_1(t), Y_2(t), \ldots, Y_{413}(t)$ the 413 functional observations constructed from the square roots of the annual arrest count for the 413 individuals in the study.

Life Course Data in Criminology V

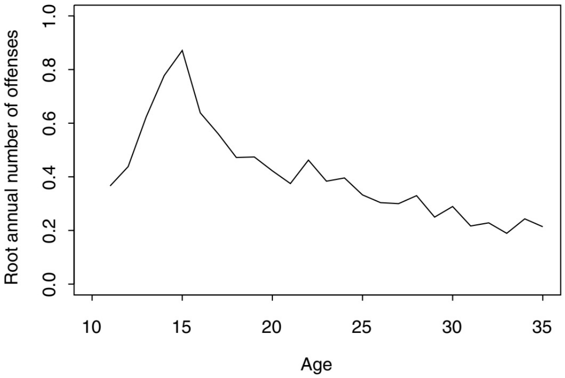


Figure 2.5. The sample mean function of the criminology functional data.

Life Course Data in Criminology VI

1. We estimate the mean

$$\bar{Y}(t) = \frac{1}{413} \sum_{i=1}^{413} Y_i(t).$$

2. It can be seen in Figure 2.5 that, despite the large number of functions on which the mean is based, there is still some fluctuation in the result of a kind that is clearly not relevant to the problem at hand; there is no reason why 29-year olds commit fewer offenses than both 28- and 30-year olds for instance!
3. We will smooth the mean function, but it should be pointed out that this particular set of data has high local variability. In many other practical examples no smoothing will be necessary.
4. There are many possible approaches to the smoothing of the curve and the one we use is a roughness penalty method.

Life Course Data in Criminology VII

5. We measure the roughness, or variability, of a curve g by the integrated squared second derivative of g. Our estimate of the overall mean is then the curve $m_\lambda(t)$ that minimizes the penalized squared error

$$S_\lambda(g) = \int \{g(t) - \bar{Y}(t)\}^2 \,\mathrm{d}t + \lambda \int \{g''(t)\}^2 \,\mathrm{d}t.$$

6. Here the smoothing parameter $\lambda \geq 0$ controls the trade-off between closeness of fit to the average of the data, as measured by the first integral in the formula and the variability of the curve, as measured by the second integral.
7. If $\lambda = 0$ then the curve $m_\lambda(t)$ is equal to the sample mean curve $\bar{Y}(t)$.
8. As λ increases, the curve $m_\lambda(t)$ gets closer to the standard linear regression fit to the values of $\bar{Y}(t)$.

Life Course Data in Criminology VIII

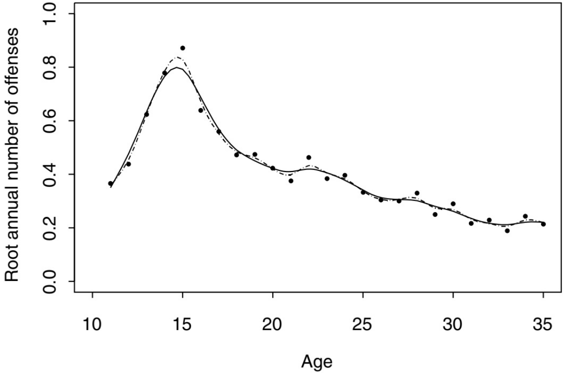


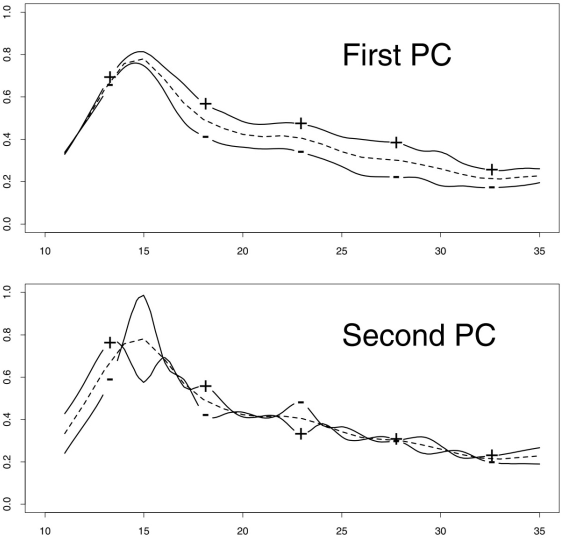
Figure 2.6. Estimate of the overall mean of the square root of the number of arrests per year. Points: raw means of the data. Dashed curve: roughness penalty smooth, $\lambda = 2 \times 10^{-7}$, cross-validation choice. Solid curve: roughness penalty smooth, $\lambda = 10^{-6}$, subjective adjustment.

Life Course Data in Criminology IX

Functional principal component analysis

1. What are the types of variability between the boys in the sample?
2. There is controversy among criminologists as to whether there are distinct criminal groups or types.
3. Some maintain that there are, for instance, specific groups of high offenders, or persistent offenders. Others reject this notion and consider that there is a continuum of levels and types of offending.

Life Course Data in Criminology X



Life Course Data in Criminology XI

1. In Figure 2.7, for each of the first three principal components, three curves are plotted. The dashed curve is the overall smoothed mean, which is the same in all cases. The other two curves show the effect of adding and subtracting a suitable multiple of the principal component weight function.
2. It can be seen that the first principal component corresponds to the overall level of offending from about age 15 to age 35.
3. All the components have a considerable amount of local variability, and in the case of the second component, particularly, this almost overwhelms any systematic effect.
4. Clearly some smoothing is appropriate, not surprisingly given the high variability of the data.

Life Course Data in Criminology XII

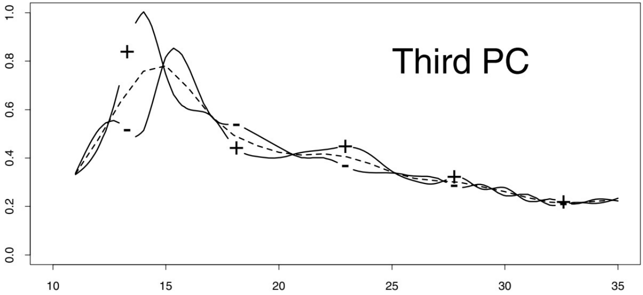
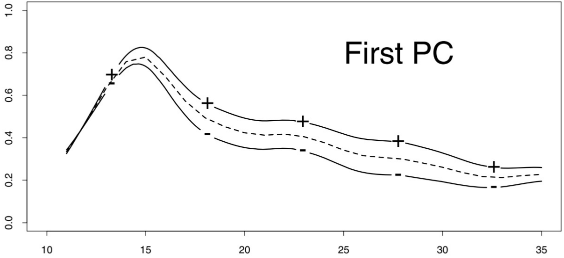


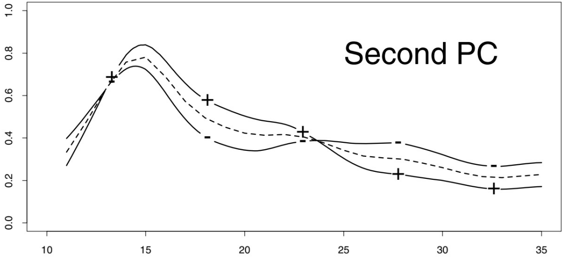
Figure 2.7. The effect of the first three unsmoothed principal components of the criminology data. In each graph, the dashed curve is the overall mean, and the solid curves are the mean $\pm$ a suitable multiple of the relevant principal component weight function. The $+$ and $-$ signs show which curve is which.

Life Course Data in Criminology XIII

1. Smoothing a functional principal component analysis is not just a matter of smoothing the components produced by a standard PCA.
2. To obtain a smoothed functional PCA, we take account of the need not only to control the size of principal component weight function, but also to control its roughness.

Life Course Data in Criminology XIV





Life Course Data in Criminology XV

1. The first three principal component weight functions arising from a smoothed PCA are given in Figure 2.8. The smoothing parameter was chosen by subjective adjustment to the value $\alpha = 10^{-5}$. It can be seen that each of these components now has a clear interpretation.

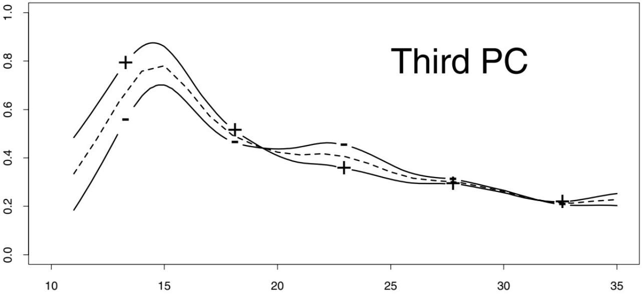


Figure 2.8. The effect on the mean curve of adding and subtracting a multiple of each of the first three smoothed functional principal components. The smoothing parameter was set to $\alpha = 10^{-5}$.

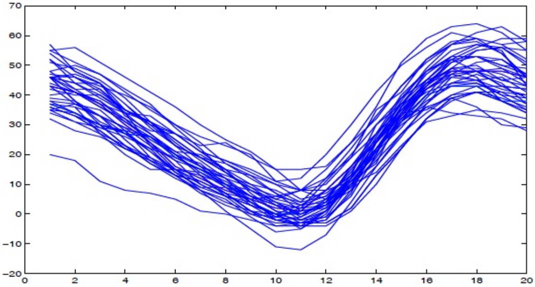
Life Course Data in Criminology XVI

1. **The first** quantifies the general level of criminal activity throughout later adolescence and adulthood. A high scorer on this component would show especially above-average activity in the years from age 18 to age 30. It is interesting that this increased difference is not in the teenage years when the general level is very high anyway. High scorers on this component are above average during late adolescence but not markedly so; it is in their late teens and twenties that they depart most strongly from the mean. For this reason we call this component "Adult crime level."
2. **The second component** indicates a mode of variability corresponding to high activity up to the early twenties, then reforming to better than average in later years. High scorers are juvenile delinquents who then see the error of their ways and reform permanently. On the other hand those with large negative scores are well-behaved teenagers who then later take

up a life of crime. We call this component "Long-term desistance."

3. **The third component** measures activity earlier in life. High scorers on this component are high offenders right from childhood through their teenage years. The component then shows a bounceback in the early twenties, later reverting to overall average behavior. This component is most affected by juvenile criminal activity and we call it "Juvenile crime level".

Functional Depth



which one is the deepest function in this cloud of functions?

- The depth is a concept which measures how deep (or central) is a datum respect to a population.
- In univariate data, the median would be the deepest point of clouds of points.

- Recall: if $x_1, \ldots, x_n$ is a sample drawn from a random variable with cumulative distribution function F, the halfspace and the simplicial depths of x_i with respect to the sample $x_1, \ldots, x_n$, are given by

$$HD_n(x_i) = \min\{F_n(x_i), 1 - F_n(x_i)\},$$

and,

$$SD_n(x_i) = 2F_n(x_i)(1 - F_n(x_i)),$$

respectively, where F_n is the empirical cumulative distribution function of the sample $x_1, \ldots, x_n$:

$$F_n(x) = \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}\{x_k \leq x\}, \quad x \in \mathbb{R}.$$

- Both depths provide very similar ordered samples $x_{(1)}, \ldots, x_{(n)}$, such that $x_{(1)}$ is the deepest point and $x_{(n)}$ is the less deepest one.

The aim of data depth for functional data is to measure the centrality of a given curve, x_i, within a set of curves, $x_1, \ldots, x_n$.

Depth in functional context provides a method to order sample curves according to decreasing depth values,

- $x_{[1]}$: the deepest (most central or median) curve,
- $x_{[n]}$: the most outlying (least representative) curve,
- $x_{[1]}, \ldots, x_{[n]}$: start from the center outwards.

The Fraiman–Muniz depth

- For every $t \in [0,1]$, let $F_{n,t}$ be the empirical distribution of the sample $x_1(t), \ldots, x_n(t)$ and let $D_n(x_i(t))$ denote the univariate depth of the data $x_i(t)$ in this sample $x_1(t), \ldots, x_n(t)$, given by

$$D_n(x_i(t)) = 1 - |1/2 - F_{n,t}(x_i(t))|.$$

- Then, the Fraiman-Muniz functional depth, hereafter FMD, of a curve x_i with respect the set $x_1, \ldots, x_n$ is given by

$$FMD_n(x_i) = \int_{\mathcal{T}} D_n(x_i(t))dt.$$

- In many cases, the curves are observed at a discretized set of different time points $0 \leq t_1 < \cdots < t_m \leq T$, and, consequently, a functional dataset of n functional curves, $x_1, \ldots, x_n$, observed at a grid of points, $t_1, \ldots, t_m$, is given by

$$\{x_i(t_j), i = 1, \ldots, n; \quad j = 1, \ldots, m\}.$$

- If this is not the case, note that we can always use some interpolation method to obtain a transformed functional dataset observed at the same grid of points.
- The sample FMD are obtained as follows:

$$SFMD_n(x_i) = \sum_{j=2}^{m} \Delta_j \Big[1 - \Big|\frac{1}{2} - F_{n,t_j}(x_i(t_j))\Big|\Big],$$

$i = 1, \ldots, n$, where $\Delta_j = (t_j - t_{j-1})$ and $F_{n,t_j}(\cdot)$ is the empirical cumulative distribution function of the points $x_1(t_j), \ldots, x_n(t_j)$.

The h-modal depth

- The h-modal functional depth, hereafter MD, of a curve x_i, with respect the set of curves $x_1, \ldots, x_n$, is given by

$$MD_n(x_i, h) = \sum_{k=1}^{n} K\Big(\frac{d(x_i, x_k)}{h}\Big),$$

where either

$$d^2(x_i, x_j) = \int_{\mathcal{T}} (x_i(t) - x_k(t))^2 dt$$

or

$$d(x_i, x_k) = \sup_{t \in \mathcal{T}} |x_i(t) - x_k(t)|,$$

K is a kernel function, and h is a bandwidth.

- Usually the truncated Gaussian kernel:

$$K(t) = \frac{2}{\sqrt{2\pi}} \exp\{-t^2/2\}, \quad t > 0,$$

is used.

- The bandwidth taken is the 15th percentile of the empirical distribution of $\{d(x_i, x_k), i, k = 1, \ldots, n\}$.
- The bandwidth h does not need to provide a very good fit of the density because we are not directly concerned here with density estimation but with support estimation. Thus, the interest is in the values around the center of the distribution, which are not very sensitive to the choice of the bandwidth.
- In fact, a wide range of values of the bandwidth are appropriate with the only requirement that the bandwidth is not very small.

The random projection depth

- The basic idea is to project each functional curve and its first derivative along a random direction, defining a point in $\mathbb{R}^2$.
- Now, a data depth in $\mathbb{R}^2$ provides an order of the projected points.
- If we use a large number of random projections, the mean value of the depths of the projected points defines a depth for functional data.

- Given the set of curves $x_1, \ldots, x_n$ and a direction v that belongs to an independent direction process $V(\cdot)$, $T_{i,v}$ is the projection of x_i along the direction v, i.e.:

$$T_{i,v} = \int_{\mathcal{T}} v(t) x_i(t) dt.$$

- Similarly,

$$T'_{i,v} = \int_{\mathcal{T}} v(t) x'_i(t) dt$$

  is the projection of the derivative of x_i, x'_i, along the direction v.
- The pair $(T_{i,v}, T'_{i,v})$ is a point in $\mathbb{R}^2$.

- Now, if $v_1, \ldots, v_p$ are p independent random directions, the random projection depth of a curve x_i is defined by

$$RPD_n(x_i) = \frac{1}{p} \sum_{k=1}^{p} D_n((T_{i,v_k}, T'_{i,v_k}))),$$

where $\boldsymbol{D_n}$ is any depth defined in R^2 of the point $(T_{i,v_k}, T'_{i,v_k}) \in \mathbb{R}^2$.
- The random directions can be generated by a Gaussian process for example.

- Let $D_n(x_1), \ldots, D_n(x_n)$ be the functional depths of the set of trajectories $x_1, \ldots, x_n$ with any of the three functional depths introduced before.
- The largest the value of $D_n(x_i)$ the deepest is the curve x_i, among the set of curves $x_1, \ldots, x_n$.
- Once the curves are ranked according to decreasing values of their depths, we get the ordered curves $x_{(1)}, \ldots, x_{(n)}$, such that $x_{(1)}$ is the deepest curve and $x_{(n)}$ is the less deepest one.
- A *functional mode* is defined as the curve that attains the maximum value of the MD depth.

- *Functional trimmed means* as the average of the most deepest $n - [\alpha n]$ curves with the FMD depth is defined as follows:

$$FTM_\alpha(x_1, \ldots, x_n) = \frac{1}{n - [\alpha n]} \sum_{i=1}^{n-[\alpha n]} x_{(i)},$$

  where α is such that $0 \leq \alpha \leq (n-1)/n$ and $[\cdot]$ denotes the integer part.

- If only one curve achieves the depth maximum value, a functional median is also defined as the functional trimmed mean with $\alpha = (n-1)/n$ and coincides with the deepest curve:

$$FMED(x_1, \ldots, x_n) = x_{(1)}.$$

- Elsewhere, the functional median is defined as the average of those curves which maximizes the depth. Note that the functional trimmed mean taking $\alpha = 0$ is the functional mean.

The band depth concept through a graph-based approach

Taken from:

Ying SUN and Marc G. GENTON. Functional Boxplots, *Journal of Computational and Graphical Statistics*, Volume 20, Number 2, Pages 316–334

- Let $y_1, \ldots, y_n$ be a sample of functions.
- The graph of a function $y(t)$ is the subset of the plane $G(x) = \{(t, y(t)) : t \in \mathcal{T}\}$.
- The band in $\mathbb{R}^2$ delimited by the curves $y_{i_1}, \ldots, y_{i_k}$ is

$$B(y_{i_1}, \ldots, y_{i_k}) = \{(t, y(t)) : t \in \mathcal{T}, \min_{r=1,\ldots,k} y_{i_r}(t) \le y(t) \le \max_{r=1,\ldots,k} y_{i_r}(t)\}$$

and constitute of functions whose graphs belong to the area delimited by the graphs of $y_{i_1}, y_{i_2}, \ldots, y_{i_k}$.

- The sample band depth of a curve $y(t)$ is

$$BD_{n,J}(y) = \sum_{j=2}^{J} BD_n^{(j)}(y),$$

where

$$BD_n^{(j)}(y) = \binom{n}{j}^{-1} \sum_{1 \le i_1 < i_2 < \cdots < i_j \le n} \mathbf{1}\{G(x) \subseteq B(y_{i_1}, \ldots, y_{i_j})\},$$

where $\mathbf{1}\{\cdot\}$ denotes the indicator function.
- The implication is that by computing the fraction of the bands containing the curve $y(t)$, the bigger the value of band depth, the more central position the curve has.

- The modified band depth (MBD, measure the proportion of time that a curve $y(t)$ is in the band:

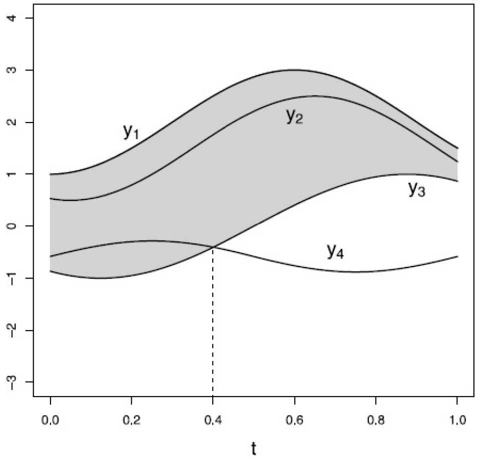
$$MBD_n^{(j)}(y) = \binom{n}{j}^{-1} \sum_{1 \le i_1 < i_2 < \cdots < i_j \le n} \lambda_r \{A(y; y_{i1}, \ldots, y_{i_j})\},$$

where

$$\begin{aligned} A_j(y) &= A(y; y_{i_1}, \ldots, y_{i_j}) \\ &:= \{t \in \mathcal{T} : \min_{r=i_1,\ldots,i_j} y_r(t) \le y(t) \le \max_{r=i_1,\ldots,i_j} y_r(t)\} \end{aligned}$$

and $\lambda_r(y) = \lambda(A_j(y))/\lambda(\mathcal{T})$, if λ is the Lebesgue measure on $\mathcal{T}$.

- If $y(t)$ is always inside the band, the modified band depth degenerates to the band depth.



An example of BD and MBD computation: the gray area is the band delimited by $y_1(t)$ and $y_3(t)$. The curve $y_2(t)$ completely belongs to the band, but $y_4(t)$ only partly does.

- Because the modified band depth takes the proportion of times that a curve is in the band into account, it avoids having too many depth ties and is more convenient to obtain the most representative curves in terms of magnitude.
- The band depth is more dependent on the shape of curves often yielding ties, thus it can be used to obtain the most representative curves in terms of shape.
- Consequently, there are two types of outliers: magnitude outliers and shape outliers.
- In general, magnitude outliers are distant from the mean and shape outliers have a pattern different from the other curves.
- A sample median function is a curve from the sample with largest depth value, defined by $argmax_{y \in y_1, \ldots, y_n} BD_{n,J}(y)$.
- If there are ties, the median will be the average of the curves maximizing depth.

- Figure above provides a simple example with $n = 4$ curves on how to compute BD and MBD in practice.
- When $J = 2$, there are six possible bands delimited by two curves. For instance, the gray area in Figure is the band delimited by $y_1(t)$ and $y_3(t)$. We can see that the curve $y_2(t)$ completely belongs to the band, but $y_4(t)$ only partly does.
- We define that a curve is contained in a band even if this curve is on the border of the band. Then $BD(y_2) = 5/6 = 0.83$ since only the band delimited by $y_3(t)$ and $y_4(t)$ does not completely contain the curve $y_2(t)$ and $BD(y_4) = 3/6 = 0.5$ as it is only completely contained in the bands delimited by itself and another curve.
- Similarly, we could compute $BD(y_1) = 0.5$ and $BD(y_3) = 0.5$.

- To compute MBD, note that the curve $y_2(t)$ is always contained in the five bands, hence $MBD(y_2) = 0.83$, the same value as BD.
- In contrast, the curve $y_4(t)$ only belongs to the band in gray 40% of the time, thus $MBD(y_4) = (3 + 0.4 + 0.4)/6 = 0.63$ by definition.
- For the other two curves, $MBD(y_1) = 0.5$ and $MBD(y_3) = 0.7$.

Functional Outlier Detection

- Outliers in a functional dataset can arise for, at least, two reasons:
  (1) outliers may be curves with gross errors such as measurement, recording, and typing mistakes. These errors should be identified and corrected whenever possible.
  (2) outliers may be real data curves in the sense that they are not gross errors but are somehow suspicious or surprising as they do not follow the same pattern as that of the rest of curves.
- We are interested in detecting and examining precisely such surprising curves as, first, they may bias our functional estimates and we would like to prevent this, and, second, they may allow to discover which sources produce these outlying curves.

- A rigorous definition of outlier in functional settings has not been given.
- In what follows, we consider that a curve is an outlier if it has been generated by a stochastic process with a different distribution than the rest of curves, which are assumed to be identically distributed.
- Therefore, we assume that the whole set of curves has been drawn from the same stochastic process, and curves that are not compatible with this assumption are outliers.
- For instance, a curve could be an outlier if it is significatively far away from the expected function of the stochastic process or have a different shape than the rest of curves.
- Note also that this definition includes the case of curves which are different from the rest only during some subintervals of the whole observation period.

- In order to identify outliers in functional datasets, we make use of functional depths.
- Depth and outlyingness are inverse notions, so that if an outlier is in the dataset, the corresponding curve will have a significant low depth.
- Therefore, a way to detect the presence of functional outliers is to look for curves with lower depths.
- Consequently, we have the following functional outlier detection procedure for detecting outliers in a given dataset of functional curves $x_1, \dots, x_n$:
  (1) Obtain the functional depths $D_n(x_1), \dots, D_n(x_n)$, for one of the functional depths, FMD, MD, or RPD.
  (2) Let $x_{i_1}, \dots, x_{i_k}$ be the k curves such that $D_n(x_{i_k}) \leq C$, for a given cutoff C. Then, assume that $x_{i_1}, \dots, x_{i_k}$ are outliers and delete them from the sample.
  (3) Then, come back to step 1 with the new dataset after deleting the outliers found in step 2. Repeat this until no more outliers are found.

- Obviously, the key point of the previous algorithm is to determinate the cut-off C, which has to be chosen to ensure a reasonable level of type I errors.
- We select C such that, in the absence of outliers, the percentage of correct observations mislabelled as outliers is approximately equal to 1%.
- In other words, we select C such that, in the absence of outliers:

$$P(D_n(x_i) \leq C) = 0.01, \quad i = 1, \ldots, n.$$

- Thus, the C taken is the 1'th percentile of the distribution of the functional depth under consideration.

- Unfortunately, the distribution of functional depths is unknown and appears to be hard to derive in a general setting.
- Thus, the cut-off C in this procedure is found by estimating this percentile, making use of the observed sample curves.
- As the sample may be contaminated by outliers, we have to be able to obtain a robust estimate of the percentile based on the sample.
- For that, one makes use of bootstrap procedure based on trimming the sample of suspicious curves, obtain smoothed bootstrap sets from the trimmed sample, and estimate the percentile based on the bootstrap sets.

The smoothed bootstrap procedure based on trimming runs as follows:

(1) Obtain the functional depths $D_n(x_1), \ldots, D_n(x_n)$, for one of the functional depths, FMD, MD, or RPD.

(2) Obtain B standard bootstrap samples of size n from the dataset of curves obtained after deleting the α% less deepest curves. The bootstrap samples are denoted by x_i^b , for $i = 1, \ldots, n$ and $b = 1, \ldots, B$.

(3) Obtain smoothed bootstrap samples $y_i^b = x_i^b + z_i^b$, where z_i^b is such that $(z_i^b(t_1), \ldots, z_i^b(t_m))$ is normally distributed with mean 0 and covariance matrix $\gamma\Sigma_x$, where Σ_x is the covariance matrix of $(x(t_1), \ldots, x(t_m))$ and γ is a bootstrap smoothing parameter. Let $y_i^b, i = 1, \ldots, n$ and $b = 1, \ldots, B$ be these samples.

(4) For each bootstrap set $b = 1, \ldots, B$, obtain C_b as the empirical 1% percentile of the distribution of the depths, $D(y_i^b)$, $i = 1, \ldots, n$.

(5) Take C as the median of the values of $C_b, b = 1, \ldots, B$.

Functional boxplots

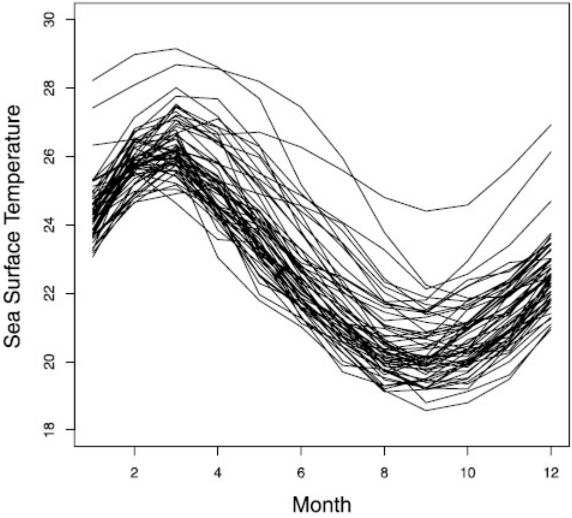
- It is well known that the boxplot is a graphical method for displaying five descriptive statistics: the median, the first and third quartiles, and the non-outlying minimum and maximum observations.
- A boxplot may also indicate which observations, if any, can be considered as outliers.
- To generalize order statistics or ranks to the multivariate setting, different versions of data depth have been introduced to measure how deep (central) or outlying an observation is.
- For functional data a notion of band depth allows for ordering a sample of curves from the center outward and, thus, introduces a measure to define functional quantiles and the centrality or outlyingness of an observation.
- Having the ranks of curves, the functional boxplot is a natural extension of the classical boxplot and is an appealing visualization tool for functional data.

Construction of functional boxplots

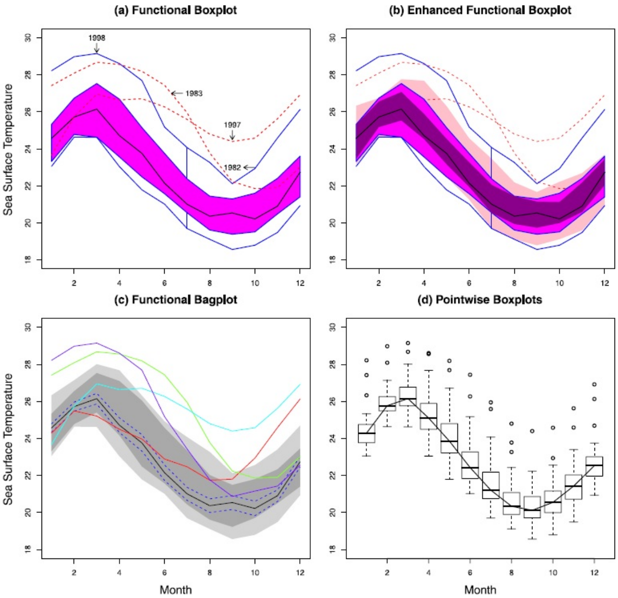
- In the classical boxplot, the box itself represents the middle 50% of the data.
- An interesting idea that can be extended to functional data is the concept of central region.
- The band delimited by the α proportion $(0 < \alpha < 1)$ of deepest curves from the sample is used to estimate the α central region.
- In particular, the sample 50% central region is

$$C_{0.5} = \left\{ (t, y(t)) : \min_{r=1,\dots,[n/2]} y_{[r]}(t) \leq y(t) \leq \max_{r=1,\dots,[n/2]} y_{[r]}(t) \right\},$$

  where $[n/2]$ is the smallest integer not less than $n/2$.
- The border of the 50% central region is defined as the envelope representing the box in a classical boxplot.
- Thus, this 50% central region is the analog to the "inter-quartile range" (IQR) and gives a useful indication of the spread of the central 50% of the curves.

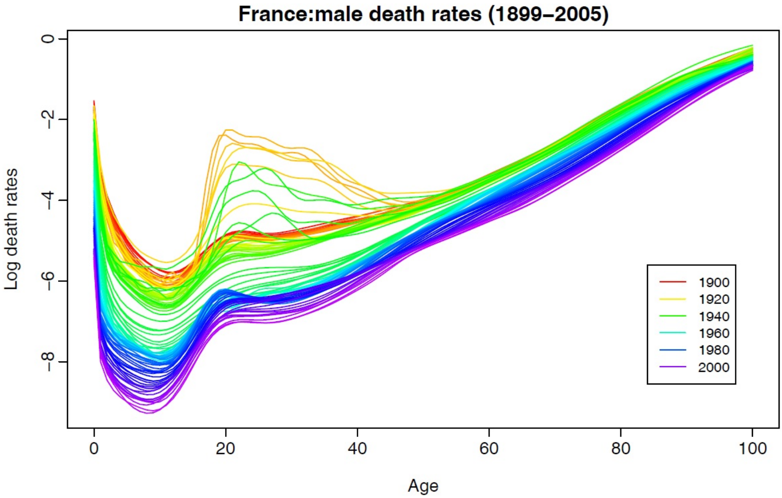


Data of monthly sea surface temperatures measured in degrees Celsius over the east-central tropical Pacific Ocean from 1951 to 2007. available online at http://www.cpc.noaa.gov/data/indices/sstoi.indices.


(a) Functional Boxplot
(b) Enhanced Functional Boxplot
(c) Functional Bagplot
(d) Pointwise Boxplots
Sea Surface Temperature
Month
1998
1983
1997
1982
30
28
26
24
22
20
18
2
4
6
8
10
12

Visualization tools

- Visualization methods help in the discovery of characteristics that might not have been apparent using mathematical models and summary statistics.
- The rainbow plot, the functional bagplot and the functional highest density region (HDR) boxplot are three graphical methods and quit new tools for visualizing large amounts of functional data in the form of smooth curves.
- This graphical method involves some kind of ordering of the functional data (time ordering, depth ordering ect.)
- The `R`-package `rainbow` for constructing rainbow plots, functional bagplots and functional HDR boxplots is available on CRAN (`http://cran.r-project.org/`).

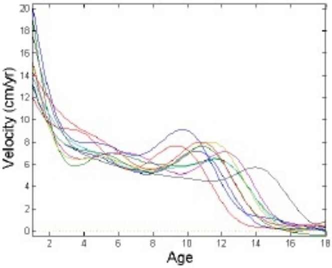


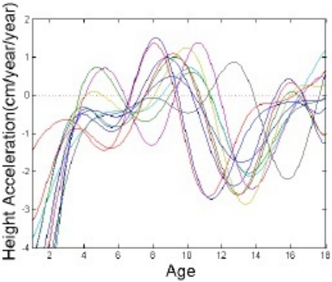
French male age-specific mortality rates (1899–2005). The oldest curves are shown in red, with the most recent curves in violet. Curves are ordered chronologically according to the colors of the rainbow.

Curves alignment

- Most analyses of functional data only account for variation in amplitude.
- Frequently, observed data exhibit features that vary in time.
- Challenge is raised by the possibility of confounding between amplitude and time variation when observed trajectories share the same underlying amplitude function, but each evolves on its own time-scale.

- For example, growth velocity curves, of the Berkeley growth study exhibit a similar growth pattern throughout, but the velocity peaks occur at different times for each child.





- A solution to this problem is to recognize time warping and to include it in the statistical modelling of functional data. Time warping can be viewed as a transformation that maps each trajectory from an internal trajectory-specific clock to absolute time.

Curve registration methods

- We assume that the observed curves $x_j(t)$ are given by

$$x_j(t) = x_j^*(h_j(t)).$$

- The sample one needs to analyse is therefore $(x_j^*(t))$, where

$$x_j^*(t) = x_j(h_j^{-1}(t))).$$

- The problem is to estimate each of $h_j(t)$ called time-warping (also registration) functions.

Landmark registration

For each curve $x_i(t)$ we choose points

$$t_{i1}, \ldots, t_{iK}$$

We need a reference (usually one of the curves)

$$t_{01}, \ldots, t_{0K}$$

so these define constraints

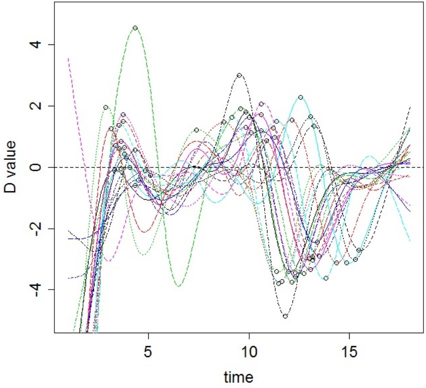
$$h_i(t_{ij}) = t_{0j}$$

Now we define a smooth function to go between these.

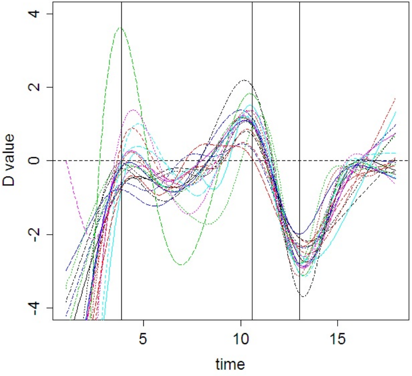
Identifying Landmarks

Major landmarks of interest:

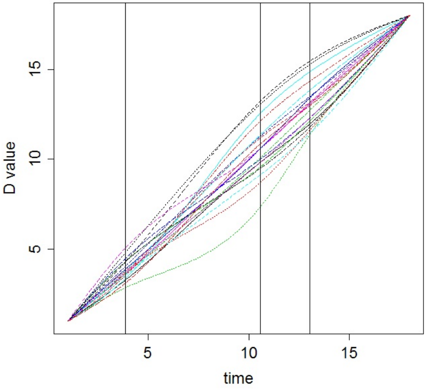
- where $x_i(t)$ crosses some value
- location of peaks or valleys
- location of infections



Results of Warping

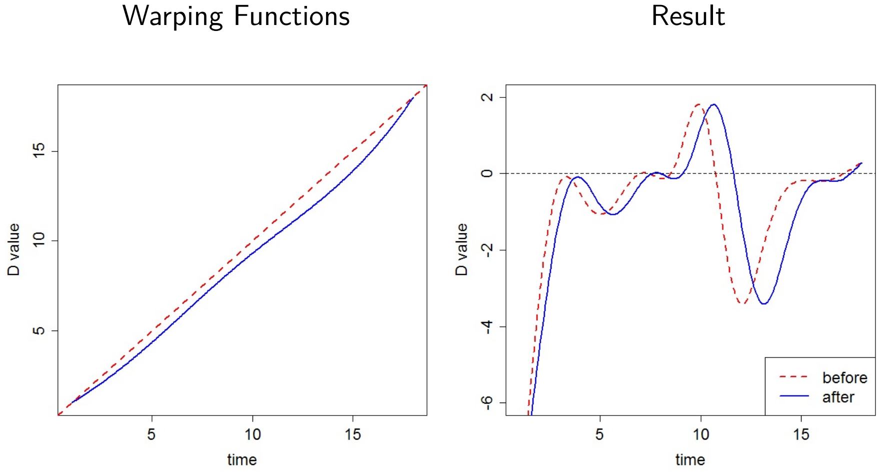


Registered Acceleration



Warping Functions

Interpretation



- Warping function below diagonal pushes registered function later in time.

Constraints on Warping Functions

Let $t \in [0 T]$, the $h_i(t)$ should follow a number of constraints:

- Initial conditions

$$h_i(0) = 0, h_i(T) = T.$$

- landmarks

$$h_i(t_{ij}) = t_{0j}$$

- Monotonicity: if $t_1 < t_2$, $h_i(t_1) < h_i(t2)$.